MBG 1201 Introduction to Modern Biology
Practice Questions

Course Lecturer: Şebnem Öztürk

seb.ozturk@gmail.com

sebnemozturk@marun.edu.tr

Practice Questions

1) Which polysaccharide is the primary form of energy storage in plants?

a) Cellulose b) Starch c) Glycogen d) Peptidoglycan

Practice Questions

2) Hemophilia A is an X-linked recessive trait. If a female who does not have hemophilia A has a son with a male who does not have hemophilia A, what is the probability that the son will not have hemophilia A if the mother's father did have hemophilia A?

a) 0% b) 15% c) 100% d) 25% e) 50%

Practice Questions

3. About a disease called alaptonuria, Garrod suggested that humans inherit two characters (alleles) for a particular enzyme and that both parents must contribute a faulty version for the offspring to have the disorder.

Is this disorder dominant or recessive, why?

Practice Questions

4. Which variable in an experiment is the one that is adjusted by the scientist?

a) dependent variable b) independent variable
c) data d) control group

Practice Questions

5. Which of the following terms refers to a supposition based on previous observations that is offered as an explanation for observed phenomenon?

a) Conclusion b) Observation
c) Theory d) Hypothesis e) Prediction

Practice Questions

6. Which of the following is not a characteristic of life?

a) homeostasis b) cellular organization

c) metabolism d) growth e) molecular organization

Practice Questions

7. What is the most important function of DNA ?

a. To carry genetic information
b. To provide the directions for protein synthesis
c. To act as a factory for protein synthesis
d. To maintain osmotic pressure

Practice Questions

8. Which of the following are macromolecules?

a. Sugars b. Proteins c. Amino acids d. Fatty acids e. Lipids

Practice Questions

9. What do all the organisms in a particular geographic area that belong to the same species form?

a) community b) population c) genus d) society

Practice Questions

10. Which of the following is not a common destination for proteins translated by ribosomes found on the endoplasmic reticulum membrane?

a) Cytosol b) Plasma membrane c) Lysosomes d) Secretory vesicles

Practice Questions

11. Which one of the following is true for the Golgi apparatus?

a. Functions in eukaryotic cells to package materials for exocytosis or for distribution to a compartment within the cell
b. Functions in prokaryotic cells to package lipids to include vesicles, which can take part in exocytosis
c. Is only present in plant cells to package seeds
d. Is present only in prokaryotic cells as a stack of flattened, saclike vesicles
e. Is found within the nucleus of eukaryotic cells

Practice Questions

12) Which of the following is a function of the nuclear envelope?

a. to separate the Synthesis of protein from synthesis of lipid
b. to separate the Synthesis of DNA and RNA from synthesis of protein and lipid
c. to separate the Synthesis of DNA from synthesis of RNA
d. to separate the Synthesis of RNA from synthesis of carbohydrate
e. to separate the Synthesis of lipid from synthesis of protein

Practice Questions

13) Which are specialized cell junctions that connect the cytoplasm of two cells and allow for the exchange of various ions and molecules?

a)Adherens junctions b) Desmosomes c) Gap junctions d) Tight junctions

Practice Questions

14) Which is a general property of cell membranes?

a) Membranes contain more lipid than protein.

b) Membrane proteins have a polar head and a non-polar tail.

c) Membrane lipids are amphipathic.

d) Membranes contain covalent bonds between fatty acids.

Practice Questions

16) Which one is true about dehydration?
a) pulls out a water from bonds
b) anabolic reaction
c) monomers come together to form a bigger polymer
d) process requires energy and is aided by enzymes
e) all of the above

Practice Questions

17) Which is true about functions of lysozymes?

a. Digestion of some food substances, damaged cells, and extracellular materials

b. Carrying out receptor-mediated endocytosis

c. Using bases to hydrolyze fats and oils

d. Pumping protons out of the cell

Practice Questions

18) Which is true about endoplasmic reticulum:

a. Not exist in eukaryotic cells

b. Sometimes found in prokaryotic cells, but, if present, is always connected to the Golgi apparatus

c. Is seldom found in prokaryotic cells but, if present, is always connected to the inner nuclear membrane

d. May have ribosomes attached, forming the rough endoplasmic reticulum, or RER

e. Accounts for only a small amount of the cellular membrane

Practice Questions

19) Which is true about the function of ribosomes

a. Conserve water and nutrients

b. Synthesize proteins from carbohydrate and lipid building blocks

c. Synthesize proteins from amino acids

d. Destroy unwanted toxic materials

e. Rid the cell of excess energy

Practice Questions

20) Which is true about hydrogen bonds

a. Are weaker than covalent bonds but stronger than ionic bonds
b. Occur between polar molecules having hydrogen atoms attached to carbon atoms
c. Are relatively unimportant, as few ions exist within cells
d. Are weaker than covalent bonds but stronger than van der Waals forces
e. Are important in chemistry but unimportant when considering living things because hydrogen bonds are toxic

Practice Questions

21) The hydrogen ion concentration of a solution having pH of 5 would is _____times higher than the hydrogen ion concentration of a solution having a pH of 9:

a. 100
b. 1,000
c. 10,000
d. 100,000
e. 1,000,000

Practice Questions

22) Which statement is true about Hydrophobic molecules:

a. Generally nonpolar and relatively insoluble in aqueous solutions
b. Generally polar and relatively insoluble in aqueous solutions
c. Generally nonpolar and relatively soluble in aqueous solutions
d. Generally polar and relatively soluble in aqueous solutions
e. Neither nonpolar nor soluble in aqueous solutions

Practice Questions

23) Why do hydrophobic substances form cohesive droplets in aqueous solutions

a. Due to Ionic bonding between water molecules

b. Due to Hydrogen bonding between water molecules, forming cages around the nonpolar droplets

c. Due to Stabilization of the hydrophobic phase by van der Waals interactions between water molecules

d. Due to The strong van der Waals forces between the nonpolar molecules

e. Due to Formation of insoluble lipid droplets

Practice Questions

24) Having two α chains and two β chains what is the maximum degree of protein structure shown by hemoglobin?

a) Primary

b)Tertiary

c) Secondary

d) Quaternary